

NEERAV BABEL

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SUMMARY

Final-year B.Tech CS student at Jain University (CGPA 9.21/10, graduating May 2027) with 1 year of freelance experience delivering 10+ production-grade full-stack and AI/ML systems for real clients across SaaS, e-commerce, and logistics. IEEE-published researcher (OTCON 2025). Seeking an entry-level Software Developer(SDE) role to drive measurable impact from day one.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Frontend: React.js, Next.js, Tailwind CSS

Backend & APIs: FastAPI, Flask, Node.js, ExpressJS, REST APIs, WebSockets, RabbitMQ, Celery, asyncio, OAuth 2.0, Stripe, Razorpay

AI/ML & GenAI: PyTorch, YOLOv11, LangChain, RAG Pipelines (Hybrid Search, Re-ranking), ChromaDB, RAGAS, scikit-learn, Hugging Face, Groq, OpenAI API, Anthropic API, Gemini API, Prompt Engineering

Cloud & DevOps: AWS, Docker, Docker Compose, Kubernetes, GitHub Actions (CI/CD), Git

Databases: PostgreSQL, MongoDB, Redis, MySQL

EDUCATION

Jain (Deemed-to-be) University — B.Tech, Computer Science & Engineering Aug 2023 – May 2027 (Expected)

CGPA: **9.21 / 10.0** | Top 5% of CSE Department | Bangalore, India

Coursework: DSA, Operating Systems, DBMS, Distributed Systems, Machine Learning, Computer Networks, OOPS, Agile/SDLC

WORK EXPERIENCE

Software Engineer (Freelance) — 10+ Clients Jun 2025 – May 2026

SaaS, E-Commerce, Logistics | *Part-time, Remote* — Concurrent with final-year studies

- Designed and shipped cloud-native REST APIs (FastAPI, Flask, Node.js) for 10+ clients with JWT auth and RBAC for secure multi-tenant systems, achieving **70% repeat engagement** and **30% faster** third-party integration timelines.
- Deployed distributed systems on AWS with Docker; maintained CI/CD pipelines via GitHub Actions ensuring zero-downtime releases, using feature-branch Git workflow with PR-based code reviews.
- Optimised PostgreSQL and MongoDB schemas with targeted indexing strategies, improving data retrieval by up to **35%**.

PROJECTS

EcoStream AI — AI Waste Management Platform Feb 2026 – Apr 2026

Team of 5 | *PyTorch, YOLOv11, FastAPI, LangChain, ChromaDB, Groq/Llama-3.1, RAGAS, React, PostgreSQL*

github.com/PrakashSh05/EcostreamAI | *My contributions:* YOLOv11 training, RAG pipeline, RAGAS benchmarking, predictive analytics

- Trained YOLOv11 (mAP50-95 = **0.6503**, +2.8pp over YOLOv8); engineered production RAG pipeline (LangChain + ChromaDB, hybrid search, re-ranking) across Groq/Llama-3.1 and OpenAI API achieving **0.92 faithfulness**, **0.86 answer relevancy** at 966ms P95 latency / 50 concurrent requests.
- Built predictive analytics engine (exponential decay, cyclical weighting, Gaussian KDE heatmaps) for next-day waste forecasting across 4 Indian cities; validated via RAGAS hybrid search evaluation.

EchoQuery — Distributed Task Engine

Oct – Nov 2025

Python, FastAPI, Redis, RabbitMQ, PostgreSQL, Kubernetes, Celery, asyncio | github.com/Neerav02/EchoQuery

- Architected fault-tolerant async task engine decoupling compute from API layer, reducing latency by **40%**; deployed on Kubernetes with auto-scaling worker pools achieving **99.99% uptime**.
- Implemented Redis distributed locking to eliminate concurrent write conflicts; engineered Dead Letter Queue with exponential backoff for zero data loss; validated via pytest suite.

Waggy — High-Concurrency E-Commerce Platform

Jan – Apr 2025

Python, Flask, Node.js, MongoDB, TypeScript, Next.js, Socket.IO, Stripe, Groq AI, OAuth 2.0 | waggy-pet-shop-ubkp.onrender.com (Live)

- Integrated Stripe and Google OAuth, increasing simulated user retention by **40%**; resolved server deadlocks by migrating to Gevent-backed WebSockets achieving <50ms updates across 100% of sessions.
- Engineered LLM chatbot (Groq, evaluated against Gemini and Anthropic APIs for latency benchmarking) and SMTP email pipeline delivering 24/7 automated support; reduced order receipt delivery to <2 seconds.

RESEARCH & PUBLICATIONS

- Crypto Seer** — ML-based financial time-series prediction. *IEEE OTCON 4.0, 2025*. doi.org/10.1109/OTCON65728.2025.11070454
- EcoStream AI** — Co-author; authored Performance Analysis and RAG Experimental Results sections. (*In submission*)

ACTIVITIES & ACHIEVEMENTS

- AtomQuest Hackathon 2026** — Top 10 Finalist (national) | **Smart India Hackathon 2025** — Institutional representative
- Agentic AI Hackathon 2026** — Built autonomous AI agent system | **HackOn With Amazon 6.0** — Cloud-native solution

CERTIFICATIONS

LangChain for LLM Application Development — DeepLearning.AI (2026), Completed with Distinction (100%)

Fundamentals of LLMs — Hugging Face (Jun 2026) | **Meta Full-Stack Developer** — Coursera (2025) | **Full Stack Web Development** — freeCodeCamp (2025)